Power Bi Project

🔹 1. Project Overview

The objective of this project was to build an **end-to-end sales analysis dashboard** using **Power BI,The data set was provided by Vidhya Analytics course**, transforming raw sales data into actionable business insights.

The project covers the complete analytics lifecycle—from **data cleaning** to **dashboard creation**.

🔹 2. Dataset description

The dataset is Super Store data in Excel format, It has three sheets 1-Orders, 2-People , 3-Return. Each sheet contains sales related columns as.

1. Orders:

- Order ID
- Order Date
- Ship date
- Ship Mode
- Customer ID
- Customer Name
- Segment
- Country
- City
- State
- Postal Code
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit

2. Returns

- Returned
- Order ID

3. People

- Person
- Region

The data was in raw format it required cleaning and transformation before analysis

🔹 3. Tools & Technologies Used

- **Power BI Desktop**
- **Power Query** (for data cleaning & transformation)
- **DAX** (basic measures and calculations)

🔹 4. Data Cleaning & Transformation (Power Query)

- The following steps were performed in Power Query:
- First load data set in the Power BI in Transform mode
- Use First row as a header tool to rearrange columns headers
- Removed null value using filter
- Created new columns using merge queries and then further column using split columns
- Renamed columns for better readability
- Created new columns using Add columns in Date option
- Save and apply
- Make relation between table/sheet

🔹 5. Dashboard design

- Design a welcome page using random website called PGP GLASS

- Copied their theme,colour scheme and font from website

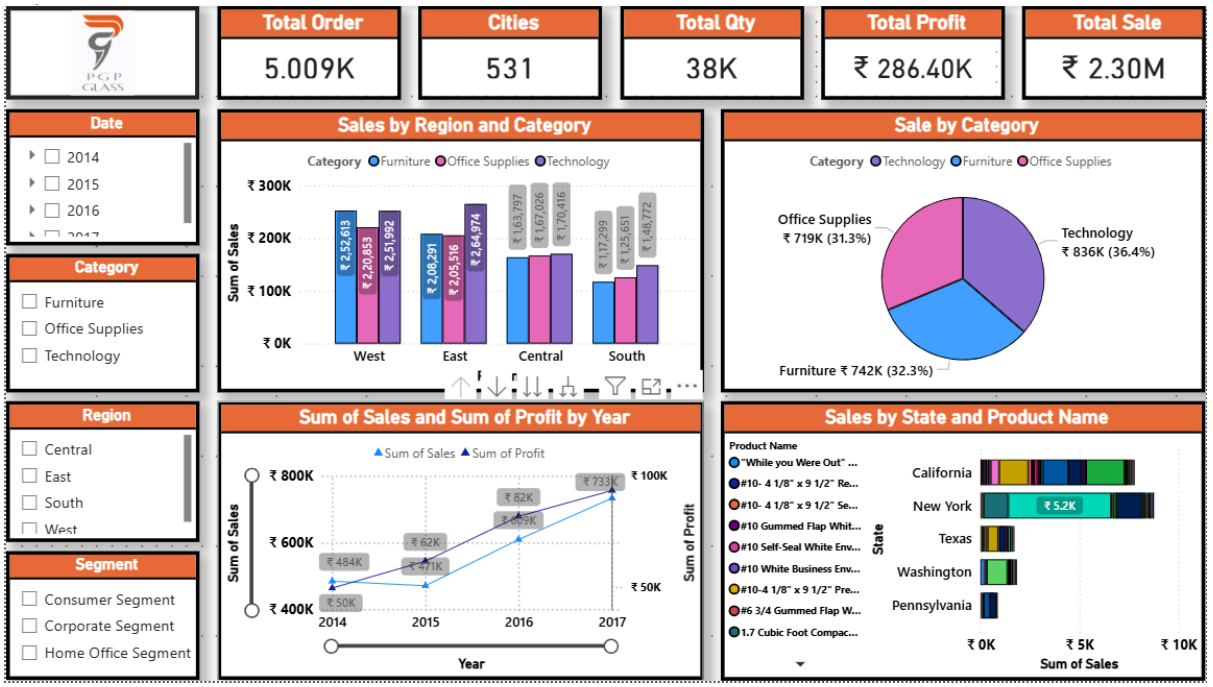
- Added 5 buttons and Navigate a pages from that:

1. Sales Overview
2. Profit Analysis
3. Time Series Analysis
4. Region Wise Analysis
5. Customer Analysis

🔹 5. Data visualization

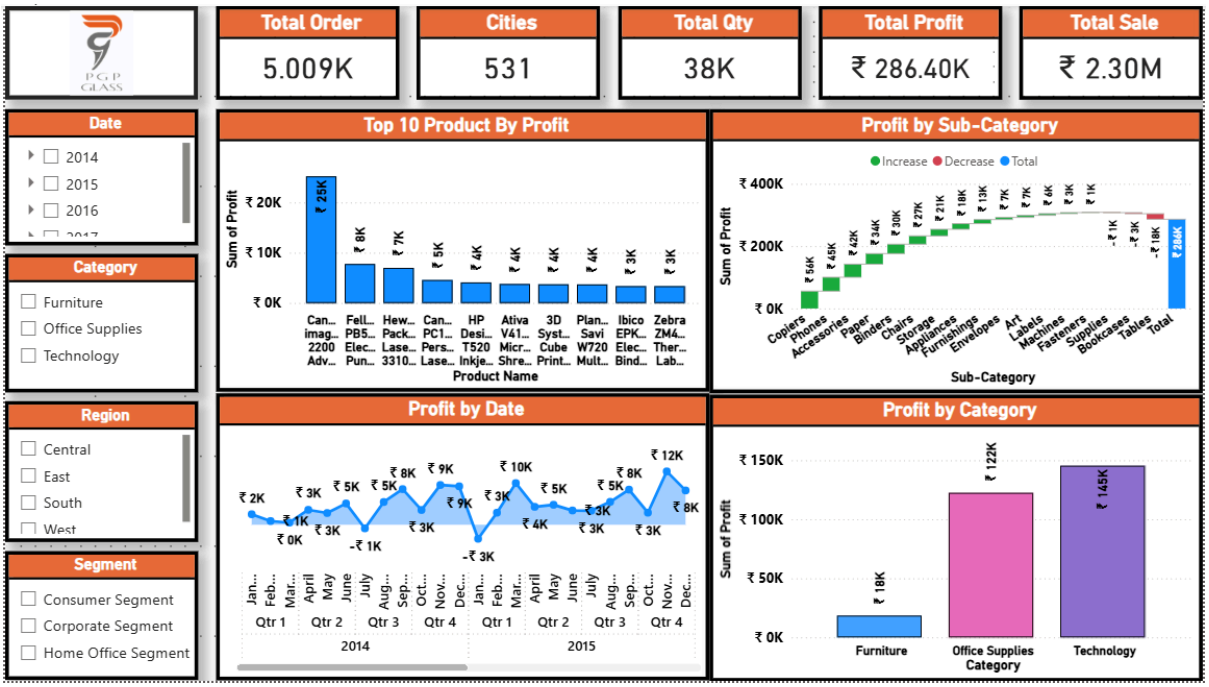
1. Sales Overview:

- Take 5 cards and add: Total Sales, Total Profit,Total QTY, Cities, Total Orders
- Add one main menu button on upper right corner using company logo
- Add 4 slicer on the left corner: Date, Category, Region, Segments
- Add 4 charts:
- Clustered column chart to measure Sales by region and category
- Add pie chart to measure Sale by category
- Add line chart to measure profit and sales by year
- Add stacked bar chart to measure sales by state and product name



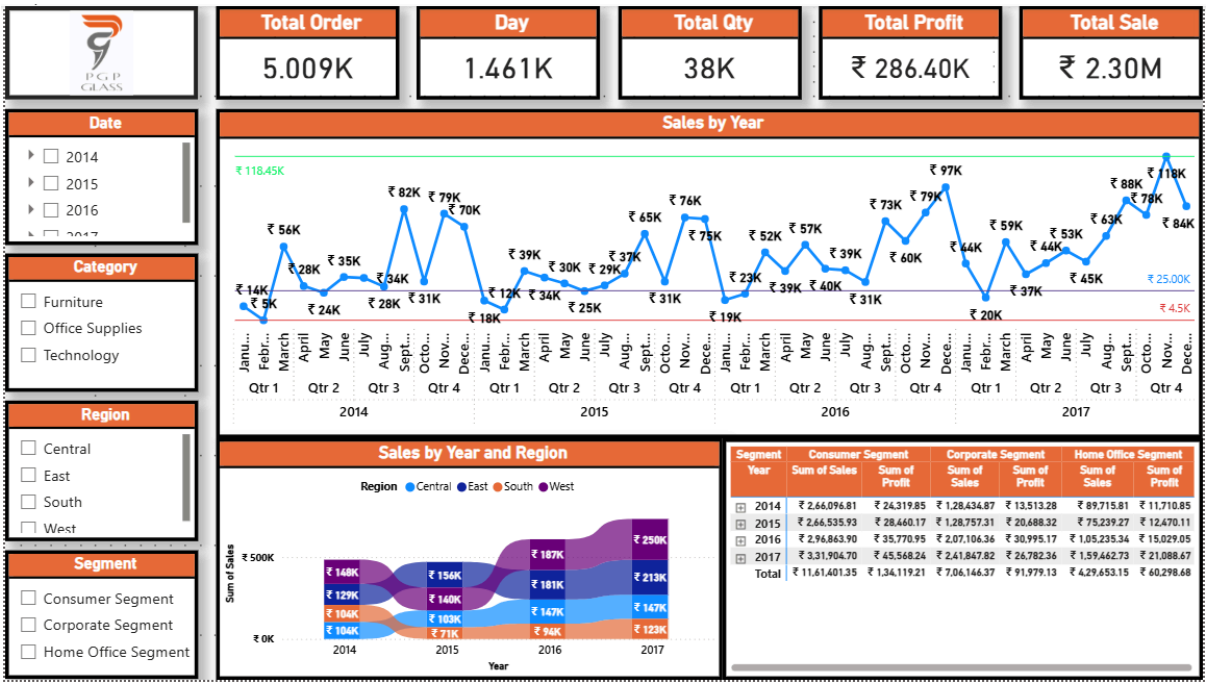
2. Profit Analysis:

- Same theme , above card, back button and 4 slicer copy from previous page
- Add 4 charts:
  A. Clustered column chart of product by Profit and add filter to 10 product only by Profit
  B. Waterfall chart to measure Profit by Sub-Category
  C. Add Area chart to measure Profit by Date and use drill down option to see quarter wise and monthly trend
  D. Take Stacked column chart add Category>Sub-Category>Product Name on X-axis and profit on Y-axis, end user can analyze using drill down option to further click on Category>Sub-Category>Product



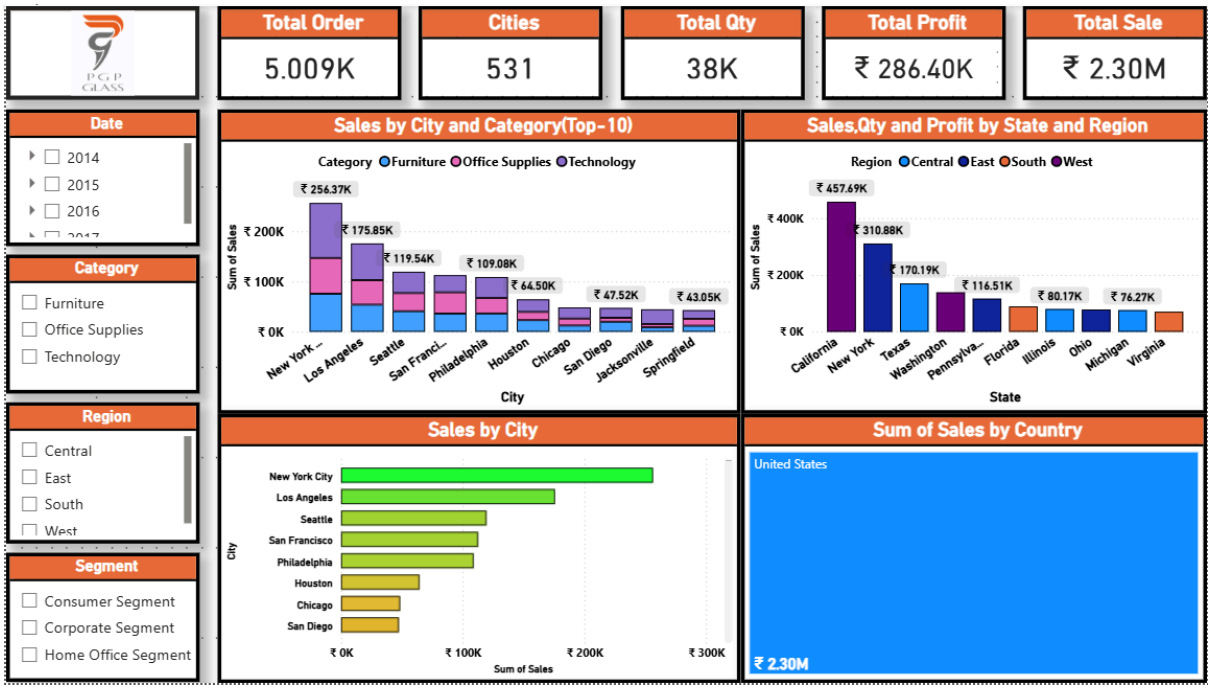
3. Time Series Analysis:

- Same theme , above card, back button and 4 slicer copy from previous page except Cities change to Day
- Add 3 charts:
    A. Add line chart to measure Sales by Year, use drill down to see monthly and quarterly trend, use further analysis and add Y-axis constant line at(25000) to analyse which month has below sales, add min line and max line
    B. Add ribbon chart to analyse sales by region and year
    C. Add matrix chart to analyse sales and profit by segments



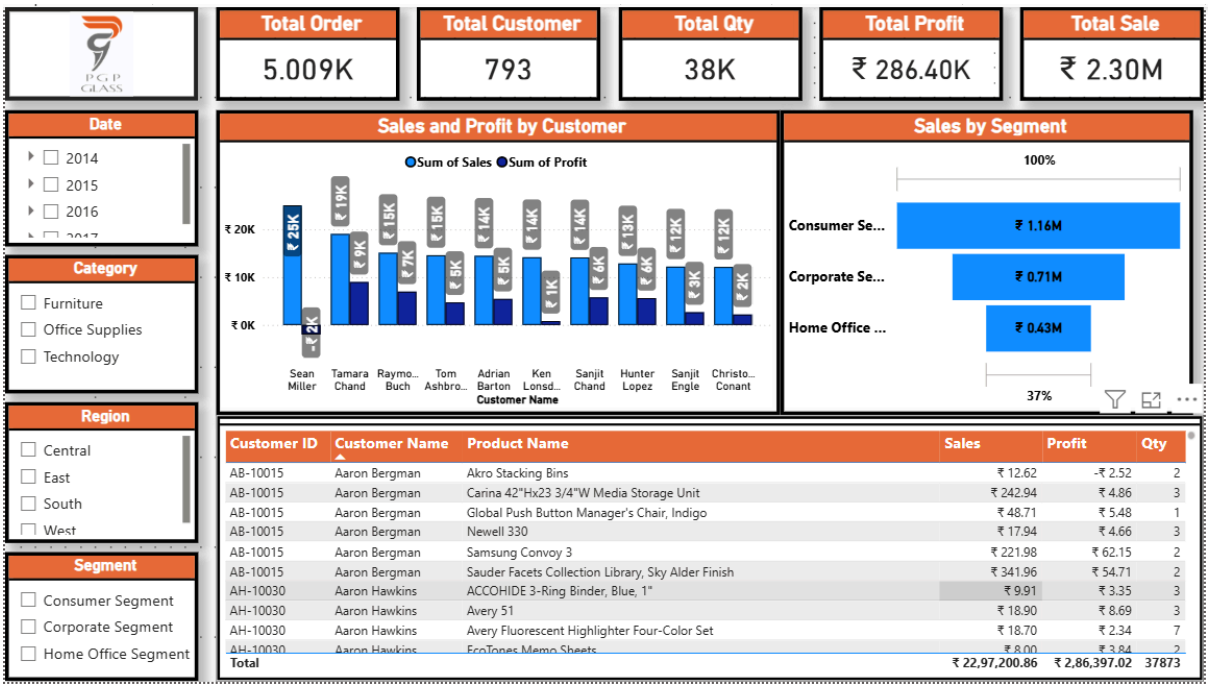
4. Region Wise Analysis:

- Same theme , above card, back button and 4 slicer copy from previous page
- Add 4 charts:
    A. Add stacked column chart to measure Sales by cities and Category, Filtered 10 Cities by Sale
    B. Add stacked column chart to measure sales by state, add Region in legend, profit and quantity in tooltips
    C. Add stacked bar chart to measure sales by cities
    D. Get Treemap and add sales in value and add hierarchy wise category Country > Region > State > City, End user can use drill down feature analyse sales likewise



5. Customer Analysis:

- Same theme , above card, back button and 4 slicer copy from previous page except city card change to total customer
- Add 3 charts:
    A. Add clustered column chart to analyze Top 10 customer by sale and profit
    B. Add Funnel chart to measure sales by segments
    C. Take Table and add Customer id, Customer name, Product name, Sale, Profit, Quantity.

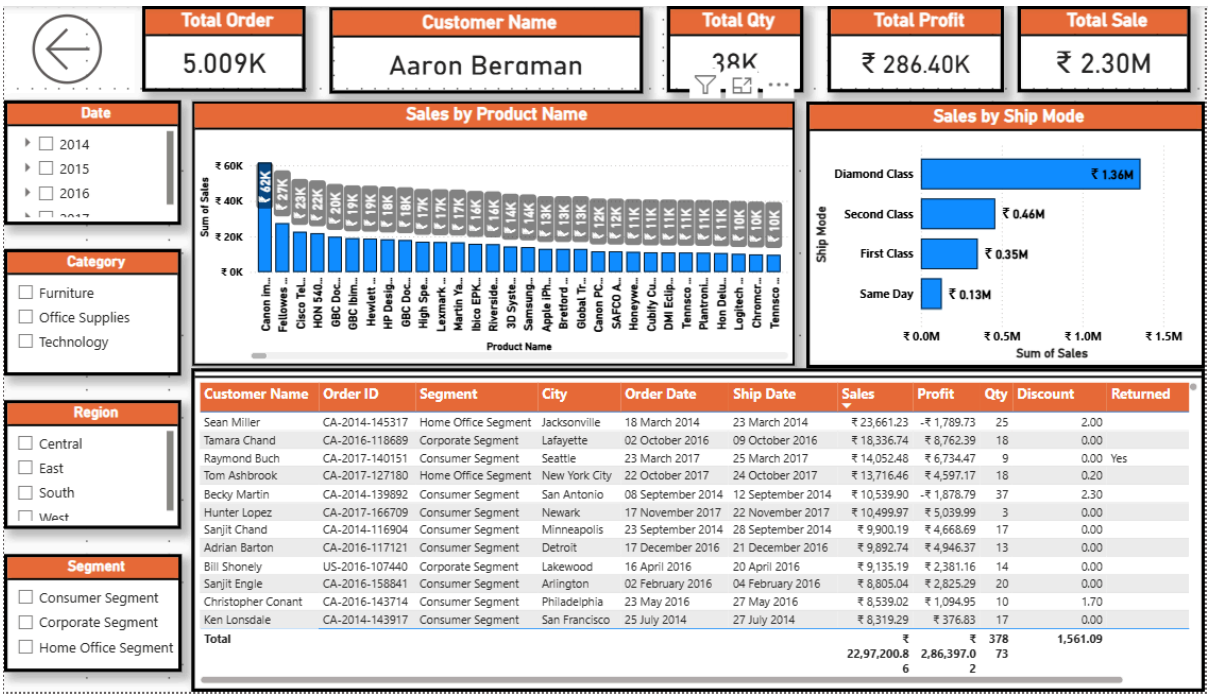


Customer ID	Customer Name	Product Name	Sales	Profit	Qty
AB-10015	Aaron Bergman	Akro Stacking Bins	₹ 12.62	-₹ 2.52	2
AB-10015	Aaron Bergman	Carina 42"Hx23 3/4"W Media Storage Unit	₹ 242.94	₹ 4.86	3
AB-10015	Aaron Bergman	Global Push Button Manager's Chair, Indigo	₹ 48.71	₹ 5.48	1
AB-10015	Aaron Bergman	Newell 330	₹ 17.94	₹ 4.66	3
AB-10015	Aaron Bergman	Samsung Convoy 3	₹ 221.98	₹ 62.15	2
AB-10015	Aaron Bergman	Sauder Facets Collection Library, Sky Alder Finish	₹ 341.96	₹ 54.71	2
AH-10030	Aaron Hawkins	ACCOHIDE 3-Ring Binder, Blue, 1"	₹ 9.91	₹ 3.35	3
AH-10030	Aaron Hawkins	Avery 51	₹ 18.90	₹ 8.69	3
AH-10030	Aaron Hawkins	Avery Fluorescent Highlighter Four-Color Set	₹ 18.70	₹ 2.34	7
AH-10030	Aaron Hawkins	EcoTones Memo Sheets	₹ 8.00	₹ 3.84	2
Total			₹ 22,97,200.86	₹ 2,86,397.02	37873

6. Detailed Customer Analysis (Hidden):

Add detailed customer analysis page but made it hidden end user can only see this page through drill through option

- Same theme , above card, back button and 4 slicer copy from previous page except total customer card change to customer name
- Add 3 charts:
  A. Add stacked column chart to analyze sale by product name
  B. Add clustered bar chart to measure sales by segments
  C. Take Table and add Customer name, Order ID, Segment, city, Order date, Ship Date, Sale, Profit, Qty, Discount, Returned.



Customer Name	Order ID	Segment	City	Order Date	Ship Date	Sales	Profit	Qty	Discount	Returned
Sean Miller	CA-2014-145317	Home Office Segment	Jacksonville	18 March 2014	23 March 2014	₹ 23,661.23	-₹ 1,789.73	25	2.00	
Tamara Chand	CA-2016-118689	Corporate Segment	Lafayette	02 October 2016	09 October 2016	₹ 18,336.74	₹ 8,762.39	18	0.00	
Raymond Buch	CA-2017-140151	Consumer Segment	Seattle	23 March 2017	25 March 2017	₹ 14,052.48	₹ 6,734.47	9	0.00	Yes
Tom Ashbrook	CA-2017-127180	Home Office Segment	New York City	22 October 2017	24 October 2017	₹ 13,716.46	₹ 4,597.17	18	0.20	
Becky Martin	CA-2014-139892	Consumer Segment	San Antonio	08 September 2014	12 September 2014	₹ 10,539.90	-₹ 1,878.79	37	2.30	
Hunter Lopez	CA-2017-166709	Consumer Segment	Newark	17 November 2017	22 November 2017	₹ 10,499.97	₹ 5,039.99	3	0.00	
Sanjit Chand	CA-2014-116904	Consumer Segment	Minneapolis	23 September 2014	28 September 2014	₹ 9,900.19	₹ 4,668.69	17	0.00	
Adrian Barton	CA-2016-117121	Consumer Segment	Detroit	17 December 2016	21 December 2016	₹ 9,892.74	₹ 4,946.37	13	0.00	
Bill Shonely	US-2016-107440	Corporate Segment	Lakewood	16 April 2016	20 April 2016	₹ 9,135.19	₹ 2,381.16	14	0.00	
Sanjit Engle	CA-2016-158841	Consumer Segment	Arlington	02 February 2016	04 February 2016	₹ 8,805.04	₹ 2,825.29	20	0.00	
Christopher Conant	CA-2016-143714	Consumer Segment	Philadelphia	23 May 2016	27 May 2016	₹ 8,539.02	₹ 1,094.95	10	1.70	
Ken Lonsdale	CA-2014-143917	Consumer Segment	San Francisco	25 July 2014	27 July 2014	₹ 8,319.29	₹ 376.83	17	0.00	
Total						₹ 22,97,200.86	₹ 2,86,397.02	378	1,561.09	

7. Findings & Business Insights:

- ### Seasonal Trend:

  Sales show clear seasonality. We observed that sales dip at the start of the year and rise toward the end of the year. For example, sales may be lowest in January-February and highest in November-December. Company can plan inventory accordingly.

- ### City Concentration (Pareto):

  A small number of Cities account for the majority of sales. This aligns with the Pareto principle noted in business: “roughly 80% of sales come from 17% Cities” . Suppose 10–12 cities are >50% of total; these high-performing cities should be prioritized. Insight: Allocate more inventory, sales effort, and promotional budget to the top 17% of cities

- ### Weekday Patterns:

  Sales vary by day of week. The time series analysis shows Wednesday, Friday, Saturday and Sunday have the highest sales and Tuesday, Thursday and Monday have the lowest, it suggests business buying patterns. Business should ensure supply chain operations are staffed for highest sales days.